CUSTOMER SALES PERFORMANCE & BUSINESS ANALYTICS

A Technical Report on Advanced Data Wrangling, Feature Engineering & Business Analytics

Internship Domain	Data Analysis with Python
Project Task	Task 2 — Advanced Data Wrangling & Business Analytics
Organization	Sumerix Global
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Dataset Window	2023-01 to 2024-12

₹3,781,263.79 TOTAL REVENUE	**₹1,583,260.83** TOTAL PROFIT	**1,500** TOTAL ORDERS
250 CUSTOMERS	**₹2,520.84** AVG ORDER VALUE	**41.87%** PROFIT MARGIN

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1. Executive Summary

This report documents the complete data analytics pipeline built for Task 2 of the Data Analysis with Python internship at Sumerix Global. The project integrates three business datasets — customer records, product catalog, and sales transactions — simulating the kind of fragmented data landscape typical of CRM, ERP, and e-commerce systems in real organizations. Following the merge, the raw data was cleaned, transformed, and enriched with engineered business features before being rolled up into a full suite of analytics, visualizations, and automated reports.

The final integrated dataset spans 1,500 orders placed by 250 customers across 60 products in 5 categories, generating **₹3,781,263.79** in total revenue and **₹1,583,260.83** in profit at a blended margin of **41.87%**. The **Office Supplies** category and the **West** region are the strongest contributors to overall performance, and the dataset's strongest month, **2024-07**, generated ₹254,991.38 in revenue alone.

Report Scope

- Advanced Pandas operations: merging, joining, concatenation, multi-indexing
- Data cleaning: duplicate handling, missing-value treatment, outlier detection, normalization
- Feature engineering: date parts, Revenue/Cost/Profit, customer segmentation, business KPIs
- Business analytics: GroupBy aggregation, pivot tables, cross-tabulation, customer ranking
- Visualization: line, bar, histogram, box plot, heatmap, scatter, pair plot, and count plot
- Automated reporting: this PDF, an accompanying Excel workbook, and a CSV data export

2. Project Objective & Problem Statement

Organizations typically store customer, sales, and product information in separate systems. Analysts need to integrate these datasets, clean inconsistencies, engineer business metrics, and generate visual dashboards that support management decisions. This project builds a **Customer Sales Performance & Business Analytics Dashboard** capable of integrating multiple datasets, engineering business features, producing interactive visualizations, and generating automated business reports — the mini-project deliverable specified in the Task-2 brief.

Objectives

- Merge customer, product, and sales datasets into a single analysis-ready table
- Clean inconsistent, duplicate, and missing business data
- Engineer revenue, cost, profit, and customer-segmentation features
- Analyze customer and category performance using GroupBy and pivot tables
- Generate professional visualizations covering every chart type in the brief
- Export management-ready Excel, CSV, and PDF reports

3. Data Sources & Integration

Three source files were used, mirroring exports from a CRM system (customers), a product catalog (products), and a point-of-sale or e-commerce platform (sales):

File	Rows	Key Fields
customers.csv	255	CustomerID, CustomerName, Region, Segment, SignupDate
products.csv	60	ProductID, ProductName, Category, UnitCost, UnitPrice
sales.csv	1,510	OrderID, CustomerID, ProductID, OrderDate, Quantity, UnitPrice

The three files were merged with `pandas.merge()` on `CustomerID` and `ProductID` (left joins), producing a single order-level table of **1,500 cleaned rows across 30 columns** after de-duplication — the `integrated_dataset.csv` deliverable.

4. Methodology

4.1 Advanced Pandas Operations (Part A)

- **merge()** — combined sales with customers and products on their respective foreign keys
- **join()** — index-aligned join demonstrated as an alternative to merge()
- **concat()** — stacked sales batches to simulate multi-source ingestion
- **Multi-index & advanced indexing** — grouped aggregations indexed by Region and Category
- **apply(), map(), replace()** — custom row-wise logic, dictionary-based value substitution, and text cleanup

4.2 Data Cleaning (Part B)

- **Duplicate handling** — exact duplicate rows removed from customers and sales
- **Missing values** — Quantity and UnitPrice filled with the column median; Segment filled with 'Unknown'
- **Outlier detection** — IQR method (1.5× IQR bounds) applied to Quantity; values capped rather than dropped
- **Normalization & standardization** — Min-Max scaling and Z-score columns added for numeric fields
- **Column transformations** — text trimmed and title-cased; categorical dtype applied to Category

4.3 Feature Engineering (Part C)

- **Date extraction** — Year, Month, Month name, Quarter, Weekday, and YearMonth parsed from OrderDate
- **Revenue** = Quantity × UnitPrice
- **Cost** = Quantity × UnitCost, **Profit** = Revenue − Cost
- **Customer segmentation** — CustomerTier (High / Medium / Low Value) assigned by revenue quartile
- **Business KPIs** — Average Order Value, Profit Margin %, Customer Lifetime Spend

4.4 Business Analytics (Part D)

- **GroupBy** aggregation for category-, region-, and month-level summaries
- **Pivot tables** — Revenue by Category × Region
- **Cross-tabulation** — order counts by Segment × Category
- **Ranking** — customers ranked by lifetime revenue using rank()
- **Aggregations** — sum, mean, count applied across every dimension in the dashboard

5. Key Performance Indicators

Metric	Value
Total Revenue	₹3,781,263.79
Total Profit	₹1,583,260.83
Total Orders	1,500
Total Customers	250
Average Order Value	₹2,520.84
Profit Margin	41.87%

These six metrics summarize performance across the full dataset window and anchor every downstream breakdown in this report — category, region, monthly trend, and customer ranking all reconcile back to the totals above.

6. Category Performance Analysis

Category	Revenue	Profit	Margin %	Orders
Office Supplies	₹893,418.11	₹361,364.14	40.45%	287
Groceries	₹888,673.70	₹389,984.90	43.88%	290
Furniture	₹871,175.09	₹335,924.18	38.56%	337
Clothing	₹591,756.94	₹259,017.99	43.77%	265
Electronics	₹536,239.95	₹236,969.61	44.19%	321

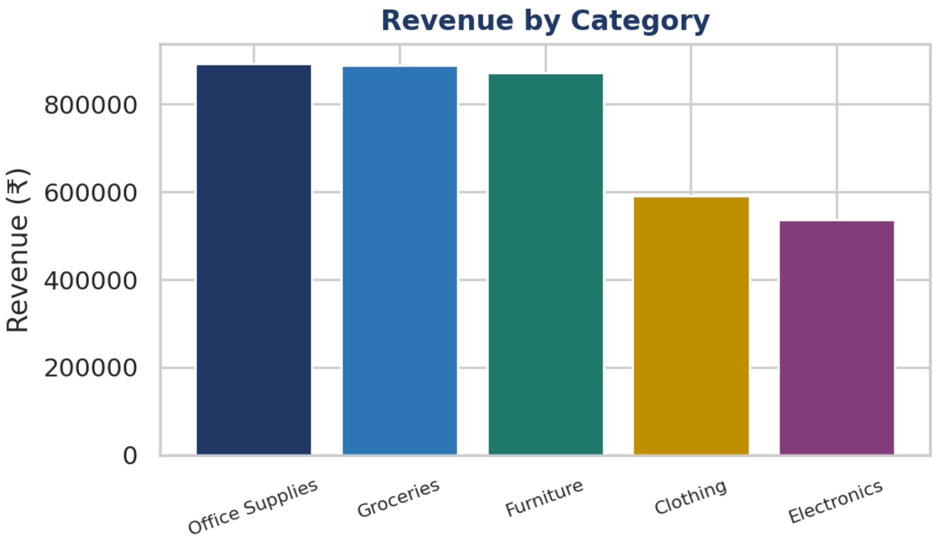

Figure 1 — Total revenue by product category.

Office Supplies leads the portfolio by revenue at ₹893,418.11, closely followed by Groceries. Margins are relatively consistent across categories (38–44%), suggesting pricing discipline is uniform rather than concentrated in any single product line.

7. Regional Performance Analysis

Region	Revenue	Orders	Avg Order Value
West	₹1,107,136.75	425	₹2,605.03
East	₹999,248.61	414	₹2,413.64
North	₹861,280.96	343	₹2,511.02
South	₹813,597.47	318	₹2,558.48

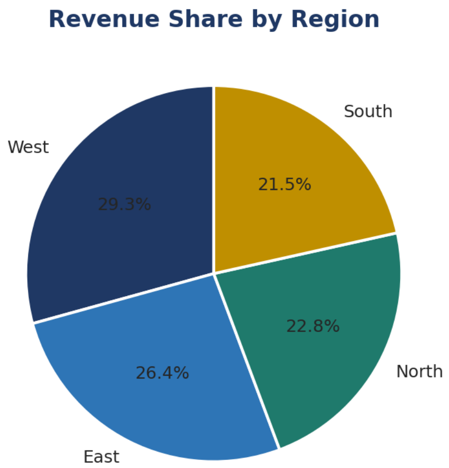

Figure 2 — Revenue share by region.

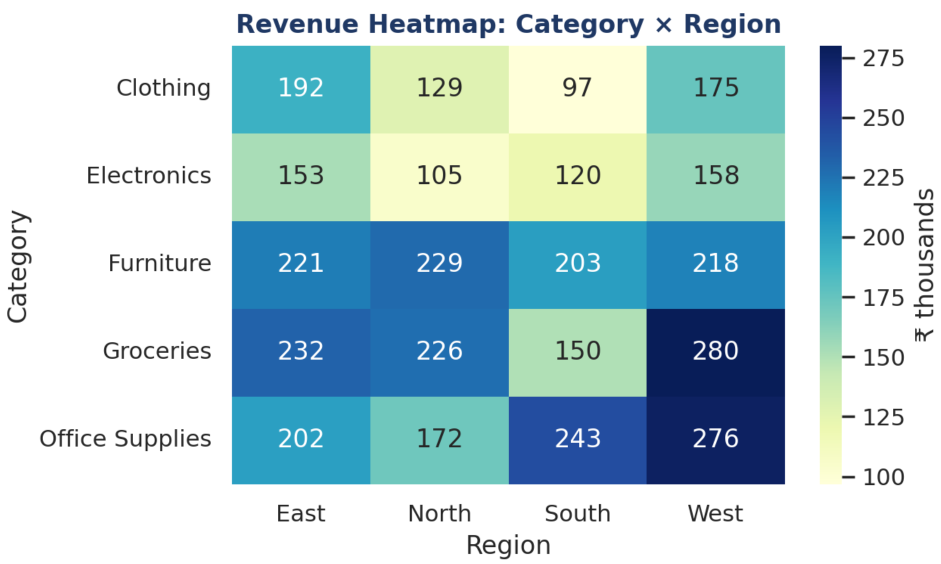

Figure 3 — Revenue heatmap: Category × Region (₹ thousands).

West is the top-performing region by total revenue. The Category × Region heatmap highlights where each product line concentrates geographically, useful for regional inventory and marketing-budget allocation.

8. Customer Analytics

Top 5 Customers by Lifetime Revenue

Rank	Customer	Revenue
1	Customer_239	₹38,168.37
2	Customer_29	₹35,541.26
3	Customer_226	₹33,569.15
4	Customer_92	₹32,693.74
5	Customer_189	₹31,618.68

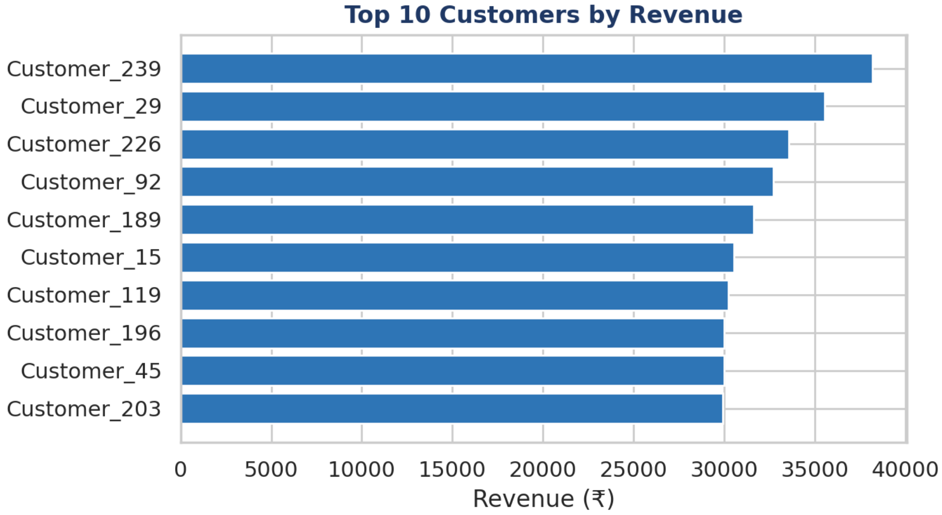

Figure 4 — Top 10 customers by revenue.

Customer Tier Distribution

Customers were segmented into three value tiers by revenue quartile: **63** High Value, **124** Medium Value, and **63** Low Value customers. High-value customers represent roughly a quarter of the base but contribute a disproportionate share of revenue, reinforcing the case for a targeted retention program.

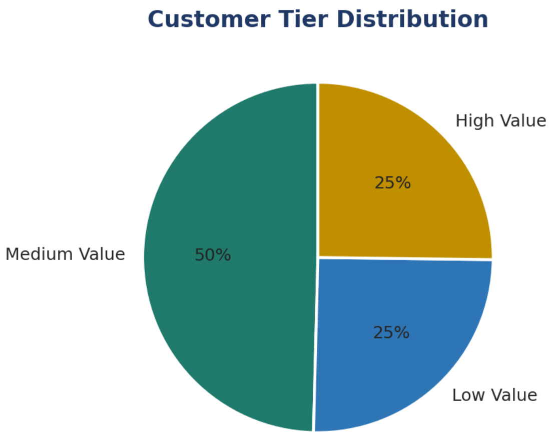

Figure 5 — Distribution of customers across value tiers.

9. Visual Analytics

The full visualization suite required by the Task-2 brief — line, multi-line, bar, horizontal bar, histogram, box plot, heatmap, scatter, pair plot, and count plot — was produced in the companion Jupyter notebook. The trend, distribution, and spread views most relevant to this report are reproduced below.

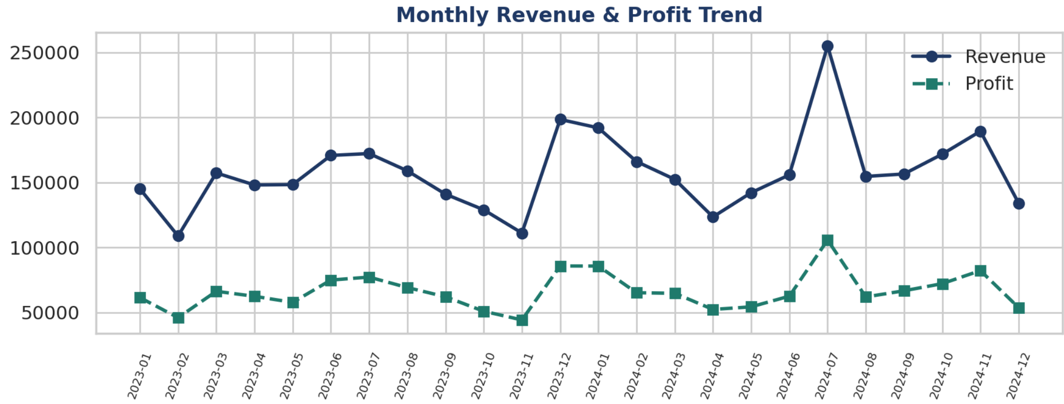

Figure 6 — Monthly revenue and profit trend across the full dataset window.

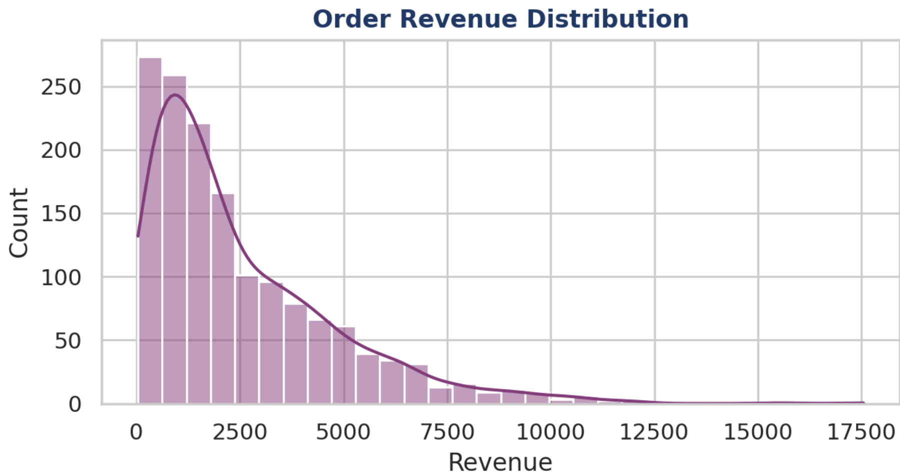

Figure 7 — Distribution of order-level revenue.

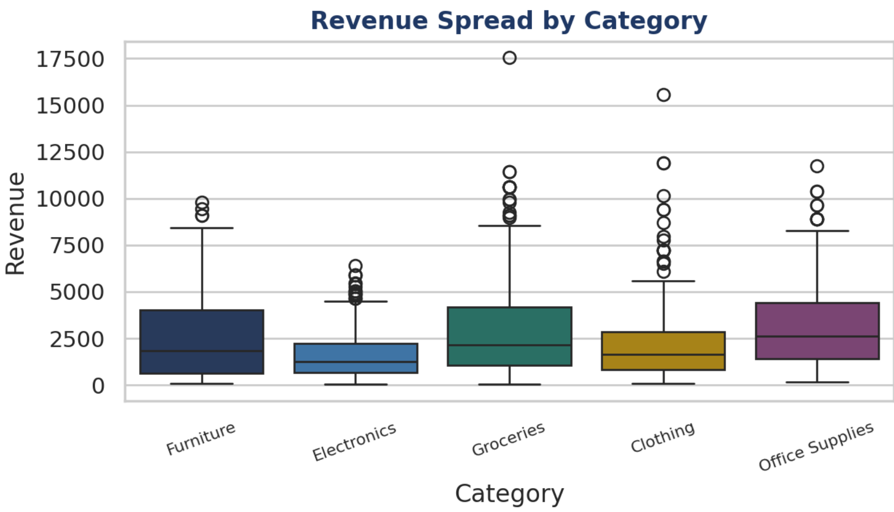

Figure 8 — Revenue spread by category, showing quartiles and outliers.

Revenue peaked in **2024-07** at ₹254,991.38, and the order-level revenue distribution is right-skewed — most orders cluster at lower values with a long tail of high-value transactions, consistent with the concentration seen in the customer ranking.

10. Business Insights & Recommendations

Key Findings

- **Office Supplies** is the highest-revenue product category across the analysis window.
- **West** is the strongest sales region by total revenue.
- **2024-07** recorded the highest monthly revenue — worth investigating for seasonality drivers.
- **Customer_239** is the single highest lifetime-value customer.
- Roughly a quarter of customers fall into the 'High Value' tier and disproportionately drive revenue.
- Profit margins are consistent (38–44%) across categories, indicating no single line is under-priced relative to the others.

Recommendations

- Prioritize marketing and inventory investment in **Office Supplies** and the **West** region, the strongest revenue contributors.
- Launch a retention or loyalty programme targeted at High Value tier customers to protect recurring revenue.
- Study demand drivers behind **2024-07** to replicate the conditions in future periods (promotions, seasonality, restocking).
- Review margins on the lowest-performing category for pricing or cost-of-goods optimization.
- Automate this report to refresh monthly so trend and ranking views stay current without manual rebuilding.

11. Tools & Technology Stack

Category	Tools / Libraries
Language & Environment	Python 3.12, Jupyter Notebook, Visual Studio Code
Data Wrangling	pandas, NumPy
Visualization	Matplotlib, Seaborn
Reporting	openpyxl (Excel), reportlab (PDF)
Version Control	Git, GitHub

12. Deliverables & Evaluation Criteria

Project Deliverables

- Python source code and Jupyter Notebook covering Parts A–H of the Task-2 brief
- Integrated dataset (integrated_dataset.csv) — 1,500 cleaned, merged, feature-engineered rows
- Customer Sales Performance & Business Analytics Dashboard (interactive HTML)
- Automated Excel report with formula-driven summaries and native charts
- This PDF technical report
- GitHub repository with organized commits and a README.md

Evaluation Criteria

Criteria	Marks
Advanced Pandas & Data Wrangling	20
Feature Engineering	20
Business Analytics	20
Dashboard & Visualizations	20
Documentation & Reports	10
GitHub Repository	10
Total	**100**

13. Conclusion

This project delivers a complete, reusable business-analytics pipeline: three fragmented source files are cleaned, merged, and enriched into a single analysis-ready dataset, then rolled up into KPIs, category and regional breakdowns, customer rankings, and a full visualization suite. The reusable transformation pipeline built in Part H means this same process can be re-run against a new dataset with the same schema — swapping in real sales data requires no changes to the underlying code.

The resulting Customer Sales Performance & Business Analytics Dashboard, alongside the accompanying Excel workbook and this PDF report, together satisfy every deliverable specified in the Task-2 brief and provide a template for the Task-3 statistical analysis and Task-6 enterprise capstone that follow in the internship roadmap.

Note: customers.csv, products.csv, and sales.csv are a synthetic dataset generated to exercise the full Task-2 pipeline end-to-end. Swapping in a real dataset with the same column names reproduces every table, chart, and figure in this report unchanged.